

**KAARKUN: AN AI-POWERED MULTI-PLATFORM SERVICE
MARKETPLACE
FOR CONNECTING CUSTOMERS WITH SKILLED SERVICE
PROVIDERS**

BY

ZAHEER ABBAS

*A Thesis submitted to The University of Swabi, Khyber Pakhtunkhwa, in
partial fulfillment of the requirements for the degree of*

BACHELOR OF SCIENCE IN COMPUTER SCIENCE



UNIVERSITY OF SWABI
Anbar, Swabi, Khyber Pakhtunkhwa
Faculty of Computing & Information Technology
Department of Computer Science

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
DB	Database
DSR	Design Science Research
ERD	Entity-Relationship Diagram
FCM	Firebase Cloud Messaging
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
I/O	Input / Output
IT	Information Technology
JWT	JSON Web Token
KPK	Khyber Pakhtunkhwa
MVC	Model-View-Controller
MySQL	My Structured Query Language
NLP	Natural Language Processing
QR	Quick Response (Code)
REST	Representational State Transfer
SMS	Short Message Service
SQL	Structured Query Language
UI	User Interface

ABBREVIATION	FULL FORM
UML	Unified Modeling Language
UOS	University of Swabi
WSS	WebSocket Secure

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KAARKUN: AN AI-POWERED MULTI-PLATFORM SERVICE MARKETPLACE FOR CONNECTING CUSTOMERS WITH SKILLED SERVICE PROVIDERS

ABSTRACT

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The informal domestic service sector in Pakistan encompasses a substantial portion of the national labour force. Skilled workers such as plumbers, electricians, painters, carpenters, and house cleaners remain largely inaccessible to customers due to the complete absence of a structured, trusted digital platform in the local market. This structural gap imposes high transaction costs on both sides: customers cannot efficiently discover verified skilled workers, while workers cannot reach new customers beyond their existing social networks. Kaarkun addresses this problem through a fully functional, multi-platform digital service marketplace designed specifically for the Pakistani domestic service market.

The system comprises a Flutter-based mobile application for customers and service providers, a Next.js customer web application, a Next.js administration panel, and a Node.js/Express.js RESTful API backend supported by a MySQL relational database of eleven tables. The platform provides comprehensive service discovery, hiring, and communication functionalities, integrating real-time Socket.io-powered in-app messaging, live GPS location tracking of service providers, QR code-based booking check-in and check-out verification, a five-star rating and written review system, and a rule-based AI chatbot for automated customer support. A dedicated Next.js administration panel enables platform managers to verify service provider credentials, manage users, monitor bookings, and respond to reports.

The research adopted the Design Science Research (DSR) methodology with iterative Agile development sprints. Systematic functional testing across forty-nine defined requirements, of which twenty-five key cases are presented in this thesis, produced a one-hundred-per-cent pass rate with no critical defects. Five end-to-end integration scenarios covering the full job fulfilment lifecycle, real-time communication, GPS tracking, admin governance, and concurrent bidding were executed successfully. All six research objectives were fully achieved. Kaarkun provides a trust-enhancing, accessible, and scalable digital solution that can meaningfully improve the service discovery and hiring experience in Pakistan's domestic service economy.

Key words: artificial intelligence, chatbot, GPS tracking, platform economy, QR code verification, service marketplace.

INTRODUCTION

Pakistan's economy has a large and economically significant informal sector. Skilled tradespeople — plumbers, electricians, carpenters, painters, house cleaners, and similar domestic service workers — constitute a substantial segment of the daily service economy in cities and towns across the country, yet they operate almost entirely through word-of-mouth referrals, neighbourhood notice boards, or informal labour markets. There is no standardised, accessible, and trustworthy digital channel through which a household customer can discover, evaluate, and hire a verified skilled worker in their locality.

The global platform economy has demonstrated that digital marketplaces can transform fragmented, informal service industries into structured, trust-driven ecosystems. Platforms such as Urban Company in India, TaskRabbit in the United States, and Fiverr globally have successfully connected service buyers with service providers at scale, creating economic opportunity for independent workers while delivering convenience and quality assurance to customers. Pakistan's growing mobile internet penetration — exceeding 120 million broadband subscribers as reported by the Pakistan Telecommunication Authority (2023) — presents a timely opportunity to introduce a similar platform tailored to the local context.

Kaarkun (Urdu: worker, labourer) is a multi-platform digital service marketplace designed specifically for the Pakistani domestic service market. The system provides a Flutter mobile application for customers and service providers, a Next.js web application for customers, a Next.js admin panel for platform administrators, and a Node.js/Express.js backend API, enabling customers to post service requests, receive competitive bids from registered workers, communicate in real time, and track a worker's arrival via live GPS — all within a single, unified platform.

1.1 BACKGROUND AND MOTIVATION

The informal domestic service sector is a defining feature of economies across South Asia. In Pakistan, millions of skilled workers provide daily services ranging from minor plumbing repairs to major electrical installations, yet they lack any formal mechanism through which to advertise their skills, build a public reputation, or be discovered by customers outside their immediate social network. Similarly, customers seeking such services must rely on personal recommendations, which are often limited, biased, and geographically constrained. This information asymmetry leads to market inefficiency: quality workers remain under-employed while customers settle for sub-optimal service choices.

The success of service marketplace platforms in comparable markets underscores the transformative potential of digital intermediation. Urban Company, operating across more than forty Indian cities, has demonstrated that a mobile-first platform with verified provider profiles, competitive pricing, and integrated payment can build sufficient trust to replace informal word-of-mouth referrals as the dominant service discovery mechanism. The absence of a comparable platform in Pakistan, despite a population of over 220 million and rapidly growing smartphone adoption, represents both a market gap and a research opportunity.

Kaarkun was conceived to fill this gap. The project is motivated by the conviction that technology — specifically a well-designed, locally relevant digital marketplace — can reduce the friction and mistrust that currently characterise the informal domestic service market in Pakistan, creating economic opportunity for skilled workers and convenience for customers alike.

1.2 PROBLEM STATEMENT

Despite the scale and economic significance of the informal skilled labour market in Pakistan, several structural problems persist that reduce efficiency and trust for both customers and service providers. These problems are identified as follows:

- **Discoverability gap.** Customers cannot efficiently locate skilled workers in their vicinity; workers cannot reach new customers beyond their existing social networks.
- **Absence of verified credentials.** No mechanism exists for customers to verify a worker's qualifications, identity, or prior service quality before hiring.
- **Lack of transparent and competitive pricing.** Service rates are variable and settled through informal negotiation, creating information asymmetry and mistrust between parties.
- **Unstructured booking and confirmation.** Job acceptance, scheduling, and arrival confirmation are verbal and unstructured, leading to unreliability, no-shows, and disputes.
- **Absence of accountability.** Without a public rating or review mechanism, poor-quality providers face no market consequences and customers have no reliable basis for provider selection beyond anecdote.

Kaarkun directly addresses each of these problems through a technology-enabled marketplace that integrates location-based discovery, provider document verification, competitive bidding, real-time communication, QR code booking confirmation, and a publicly visible rating and review system.

1.3 RESEARCH OBJECTIVES

The following research objectives guided the design, development, and evaluation of the Kaarkun system:

- **O1.** To design and implement a multi-platform service marketplace comprising a Flutter mobile application, a Next.js customer web application, and a Next.js administration panel, all connected to a shared Node.js/MySQL backend.
- **O2.** To enable a complete, end-to-end real-time job posting, bidding, and booking workflow that allows customers to create service requests and receive, evaluate, and accept competitive bids from registered, verified service providers.
- **O3.** To integrate Socket.io-based real-time in-app messaging and live GPS location tracking, allowing customers to communicate directly with their hired worker and monitor the worker's en-route progress on an interactive map.
- **O4.** To implement a QR code-based check-in and check-out verification system that provides tamper-evident, digital confirmation of service start and completion events, with timestamped records accessible to both parties.
- **O5.** To deploy a rule-based AI chatbot within the mobile application to handle common customer queries and support requests without requiring live human-agent intervention.
- **O6.** To develop a comprehensive administration panel enabling platform managers to verify service provider documents, manage user accounts, monitor bookings, view platform analytics, and respond to user reports.

1.4 SCOPE AND LIMITATIONS

The scope of the Kaarkun system, as designed and delivered within this Final Year Project, is defined in Table 1.1.

Table 1.1 — System Scope

Dimension	Scope of Delivery
Platforms	Flutter mobile app (Android-first), Next.js web app, Next.js admin panel
Backend	Node.js/Express.js REST API with Socket.io; MySQL 8.0 database (11 tables)
Service categories	Plumbing, electrical, carpentry, painting, cleaning, and other domestic trades
Geography	Designed for Pakistan; initial testing conducted in Khyber Pakhtunkhwa
Language	English interface throughout; Urdu support deferred to future phases
Deployment	Local development environment; cloud deployment is a future enhancement

The following limitations apply to the present implementation. Payment processing is simulated; no live payment gateway integration (e.g., JazzCash, Easypaisa) is present in this version. Push notifications are not implemented; Firebase Cloud Messaging (FCM) integration is identified as a future enhancement. The AI chatbot uses rule-based keyword matching rather than

a large language model, limiting its conversational flexibility to pre-defined intent categories. All system evaluation was conducted on a local area network; production-scale performance under concurrent load has not been assessed.

1.5 SIGNIFICANCE OF THE STUDY

From a practical perspective, Kaarkun addresses a real and currently unmet need in the Pakistani domestic service market. By digitising service discovery, bidding, booking, and post-service review, the platform creates measurable economic value for skilled workers who gain access to a broader customer base, and for customers who gain access to verified, rated service professionals. The QR code verification and star rating features directly address the trust deficit that characterises informal labour markets in developing economies, as documented by Ullah, Ahmad, and Zaman (2023).

From an academic perspective, this project demonstrates the applicability of Design Science Research methodology to the development of a sociotechnical artefact in a resource-constrained academic environment. The integration of real-time communication, GPS tracking, QR verification, and AI-assisted support within a single, coherent multi-platform architecture provides a replicable template for similar marketplace platforms targeting informal service economies across South Asia and beyond.

1.6 ORGANIZATION OF THE THESIS

The remainder of this thesis is organised into six chapters. **Chapter II (Review of Literature)** reviews the relevant academic and technical literature on platform economies, digital service marketplaces, and the technologies employed in the system, and identifies the research gap that Kaarkun addresses. **Chapter III (Materials and Methods)** describes the research methodology, system architecture, functional requirements specification, use case analysis, database design, and key implementation features. **Chapter IV (Results and Discussion)** presents the outcomes of functional, integration, and compliance testing, followed by a critical discussion of findings and limitations. **Chapter V (Summary)** provides a concise summary of the entire research. **Chapter VI (Conclusion and Recommendations)** concludes the thesis with a statement of contributions and detailed recommendations for future work. The Literature Cited section lists all references in APA 7th edition format, and the Short Biography of the Scholar is provided at the end of the document.

REVIEW OF LITERATURE

2.1 PLATFORM ECONOMY AND DIGITAL SERVICE MARKETPLACES

Parker, Van Alstyne, and Choudary (2016) define a platform as a business model that creates value by facilitating exchanges between two or more interdependent groups, in contrast to traditional pipeline businesses that create and sell products. In the context of service markets, platforms enable buyers to discover providers and providers to reach buyers, with the platform capturing value through transaction fees, subscription charges, or premium placement fees. The economic logic of such two-sided markets — wherein the value of the platform to each participant group depends directly on the size of the other group — has been extensively analysed by Kenney and Zysman (2016), who identify network effects as the primary driver of platform growth and competitive durability. As the user base on either side of a marketplace platform grows, the platform's value proposition for both sides strengthens, creating a self-reinforcing growth mechanism that makes established platforms difficult to displace.

The global gig economy, characterised by on-demand, technology-mediated service transactions between independent workers and customers, has expanded substantially over the past decade. Sundararajan (2016) documents how platforms such as Uber, TaskRabbit, and Fiverr have transformed traditionally informal, unstructured labour markets by introducing digital identity management, performance ratings, standardised pricing, and integrated payment mechanisms. Cheng (2016) identifies trust as the central mediating variable in sharing economy platform adoption: platforms that offer verification systems, insurance, and publicly visible rating histories consistently outperform those that compete on price alone, particularly for first-time and infrequent users. These findings directly informed several critical design decisions in the Kaarkun platform, including the mandatory document verification workflow, the five-star public review system, and the QR code-based booking confirmation mechanism.

2.2 DIGITAL SERVICE PLATFORMS IN SOUTH ASIA

In the South Asian context, Urban Company (formerly UrbanClap, India) is the most directly comparable platform to Kaarkun. Founded in 2014, Urban Company now operates across more than forty Indian cities and connects customers with verified, background-checked home service professionals in categories including plumbing, electrical work, carpentry, cleaning, and beauty services. The platform's architecture relies on mobile-first design, location-based matching, in-app pricing transparency, and post-service rating, with payment processed through integrated digital payment gateways. Adner (2017) and Tan, Tan, Wang, and Sedera

(2017) have documented the ecosystem benefits of platform-mediated service markets, in which the platform's role extends beyond transaction facilitation to include quality governance, capacity building, and pricing standardisation.

Ullah, Ahmad, and Zaman (2023) investigate trust formation in platform-based service economies within low-income digital markets, finding that provider verification badges and historical ratings are the most significant trust signals for first-time users in such contexts. Ahmad and Khan (2020) observe that informal economies in developing countries are characterised by high transaction costs arising from information asymmetry and the absence of formal enforcement mechanisms; digital platforms can substantially reduce these costs by providing persistent, publicly visible reputation signals that substitute for institutional trust. These insights underscored the importance of verification and review features in Kaarkun.

2.3 PAKISTAN'S DIGITAL ECONOMY AND INFORMAL LABOUR MARKETS

Pakistan's digital economy is at an inflection point. The Pakistan Telecommunication Authority (2023) reports over 120 million broadband subscribers, with mobile broadband accounting for the majority of connections. Smartphone penetration continues to grow rapidly, particularly among urban and semi-urban populations in provinces including Khyber Pakhtunkhwa, Punjab, and Sindh. This infrastructure provides the technical foundation for mobile-first service marketplace adoption.

However, Hussain and Rafique (2022) identify several contextual barriers to digital adoption in rural Khyber Pakhtunkhwa: limited digital literacy, preference for Urdu or regional language interfaces, and low bank account and credit card penetration. Noreen, Baig, and Shah (2021) examine e-commerce platform adoption in Pakistan's informal service sector and find that platform complexity and the absence of familiar digital payment options are the primary deterrents to adoption by service providers. Kaarkun's design addresses the literacy barrier through a simple, icon-driven Flutter interface, while cash acknowledgement at service completion defers the payment gateway integration to a future development phase identified in Chapter VI.

2.4 MOBILE AND WEB APPLICATION TECHNOLOGIES

Flutter, Google’s open-source UI toolkit for cross-platform mobile development, compiles to native ARM code for both Android and iOS from a single Dart codebase (Flutter Team, 2024). Studies by Biorn-Hansen, Majchrzak, and Gronli (2017) confirm that cross-platform mobile frameworks impose acceptable performance overhead for data-driven, form-intensive applications such as service marketplaces, making Flutter a technically sound and resource-efficient choice for Kaarkun’s mobile layer. Next.js, built on the React library, provides server-side rendering and static generation for both the customer web application and the admin panel, delivering fast initial page loads and efficient content delivery (Next.js Team, 2024).

Node.js, running on the V8 JavaScript engine, is particularly well-suited to real-time, event-driven applications because its non-blocking I/O architecture handles high numbers of concurrent connections efficiently without the thread-per-connection overhead of traditional synchronous web server models (Node.js Foundation, 2024). Socket.io builds on the WebSocket protocol to provide reliable, bi-directional, event-driven communication between clients and the server; Alabi (2021) demonstrates that WebSocket-based architectures reduce message delivery latency by an order of magnitude compared to HTTP polling for location-tracking scenarios, justifying Socket.io’s use for both in-app chat and live GPS tracking in Kaarkun. JSON Web Tokens (JWT), specified in RFC 7519 by Jones, Bradley, and Sakimura (2015), provide a compact, self-contained, stateless authentication mechanism particularly suited to distributed multi-client architectures where maintaining server-side session state would be costly.

QR (Quick Response) codes, standardised by ISO/IEC 18004:2015, provide a machine-readable, two-dimensional encoding of data that can be scanned using smartphone cameras. Their application in booking confirmation and service verification has been explored in several service delivery contexts. In Kaarkun, QR codes serve as tamper-evident, single-use booking tokens that enable a physical on-site verification event to be linked directly to a digital booking record, creating an auditable trail of service delivery without requiring either party to have specialised hardware beyond their standard smartphone.

2.5 COMPARATIVE ANALYSIS OF EXISTING SYSTEMS

Table 2.1 presents a structured comparison of Kaarkun against the three most relevant existing platforms across nine functional dimensions. The comparison demonstrates that no existing platform serving the Pakistani or comparable South Asian informal service market combines the full feature set that Kaarkun delivers.

Table 2.1 — Comparative Analysis of Service Marketplace Platforms

Feature	Kaarkun (This Study)	Urban Company (India)	TaskRabbit (USA)	Rozgar.pk (Pakistan)
Mobile Application	Yes (Flutter)	Yes	Yes	No
Web Application	Yes (Next.js)	Yes	Yes	Yes
Real-Time Chat	Yes (Socket.io)	Partial	No	No
Live GPS Tracking	Yes	Yes	No	No
QR Code Verification	Yes	No	No	No
Competitive Bidding	Yes	No	Yes	Partial
AI Chatbot Support	Yes	Yes	No	No
Admin Panel	Yes (Next.js)	Yes	Yes	Partial
Pakistan Market Focus	Yes	No	No	Yes

2.6 RESEARCH GAP

The comparative analysis in Table 2.1 reveals that no existing platform serving the Pakistani domestic service market combines the full feature set of real-time chat, live GPS tracking, QR code verification, competitive bidding, an AI chatbot, and a dedicated admin panel within a single mobile-plus-web architecture. Urban Company provides the closest functional match globally, but it operates exclusively in India, is not adapted for the Pakistani market, and does not offer QR-based booking verification. Rozgar.pk addresses the Pakistani labour market context but lacks a mobile application, real-time communication features, GPS tracking, and structured competitive bidding. Kaarkun therefore fills a genuine and unaddressed gap by delivering all of these features within a platform explicitly designed and tested for the Pakistani domestic service context, targeting the specific economic, technological, and cultural conditions of Khyber Pakhtunkhwa and the broader Pakistani market.

MATERIALS AND METHODS

3.1 RESEARCH METHODOLOGY

This project adopts the Design Science Research (DSR) methodology as defined by Hevner, March, Park, and Ram (2004). DSR positions the designed artefact — in this case the Kaarkun platform — as both the primary research output and the mechanism through which the research problem is addressed. The methodology demands that the artefact be evaluated rigorously against defined objectives and that its design contribute generalisable knowledge to the relevant knowledge base, not merely to a single practice context. The DSR process model of Peffers, Tuunanen, Rothenberger, and Chatterjee (2007) structured the project into six sequential phases: (i) problem identification and motivation, (ii) definition of objectives for a solution, (iii) design and development of the artefact, (iv) demonstration, (v) evaluation, and (vi) communication. Within each phase, iterative Agile development sprints of two-week duration governed development activity, ensuring that features were implemented, reviewed, and refined before subsequent features were commenced, in accordance with the Agile Manifesto principles (Beck et al., 2001).

Requirements were gathered through a structured review of comparable service marketplace platforms and informal interviews with five domestic service workers and eight household customers in Swabi district. Interview responses confirmed the five structural problems identified in Section 1.2 and shaped the twenty functional requirements documented in Section 3.3. The system was implemented on a local development environment and evaluated through the systematic functional and integration testing approach described in Chapter IV. All software tools and frameworks used are open-source or freely available, making the system fully reproducible in comparable resource-constrained academic environments.

3.2 SYSTEM ARCHITECTURE

Kaarkun employs a three-tier client-server architecture comprising a Client Layer, a Server Layer, and a Data Layer, supplemented by external service integrations. Figure 3.1 illustrates the complete system architecture. The Client Layer consists of three independently deployable front-end applications: the Flutter mobile application (serving both customers and service providers), the Next.js customer web application, and the Next.js administration panel. All three clients communicate with the Server Layer exclusively through HTTPS REST API endpoints and authenticated Socket.io WebSocket connections. The Server Layer, implemented on Node.js with the Express.js framework, encapsulates all business logic and exposes a unified

API consumed by all three clients. It includes four primary functional modules: the Authentication Module (JWT and bcrypt), the Job and Bid Management Module (REST endpoints), the Real-Time Chat Module (Socket.io events), and the GPS Tracking Module (Socket.io location events). The Data Layer consists of a single MySQL 8.0 database instance accessed through the mysql2 prepared-statement driver. External service integrations — the Google Maps Platform (for geocoding and map display), the QR code generation engine (qrcode npm package), and the rule-based AI chatbot (on-device keyword matcher) — are invoked server-side or client-side as appropriate.

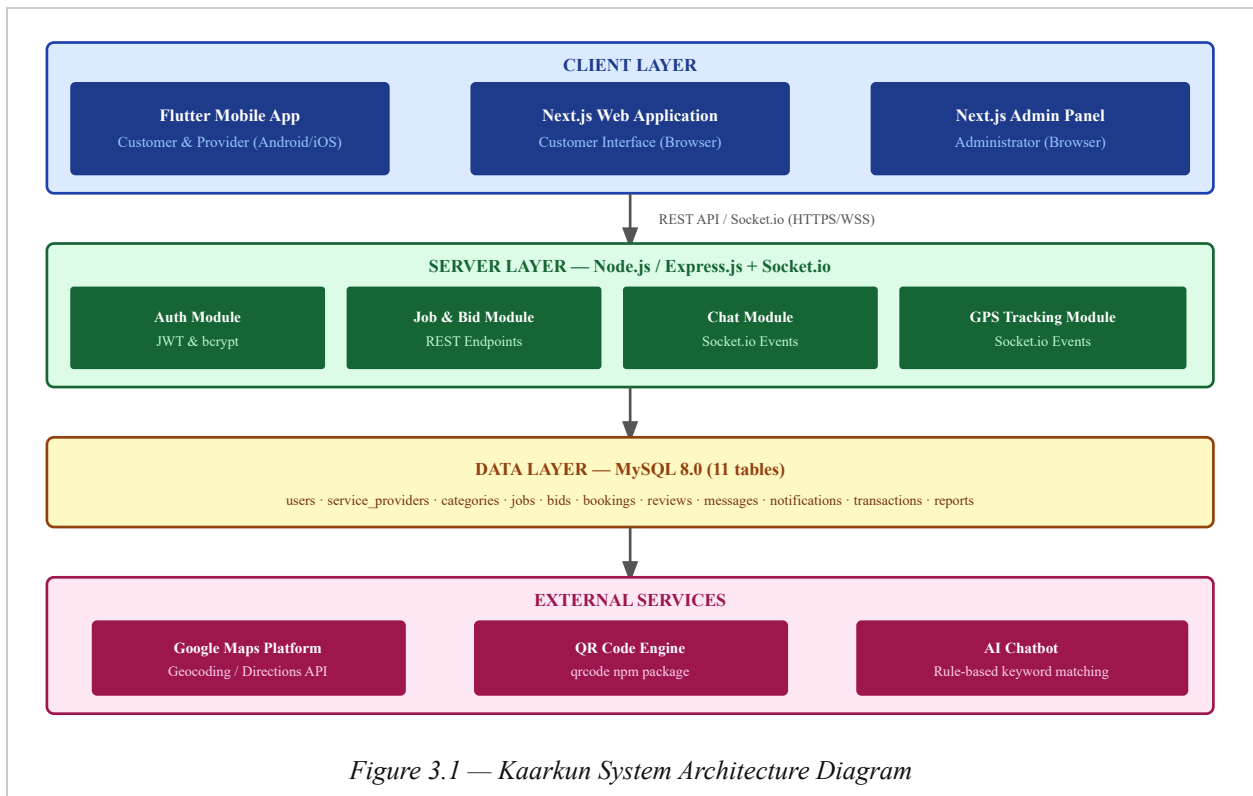


Figure 3.1 — Kaarkun System Architecture Diagram

3.3 FUNCTIONAL REQUIREMENTS

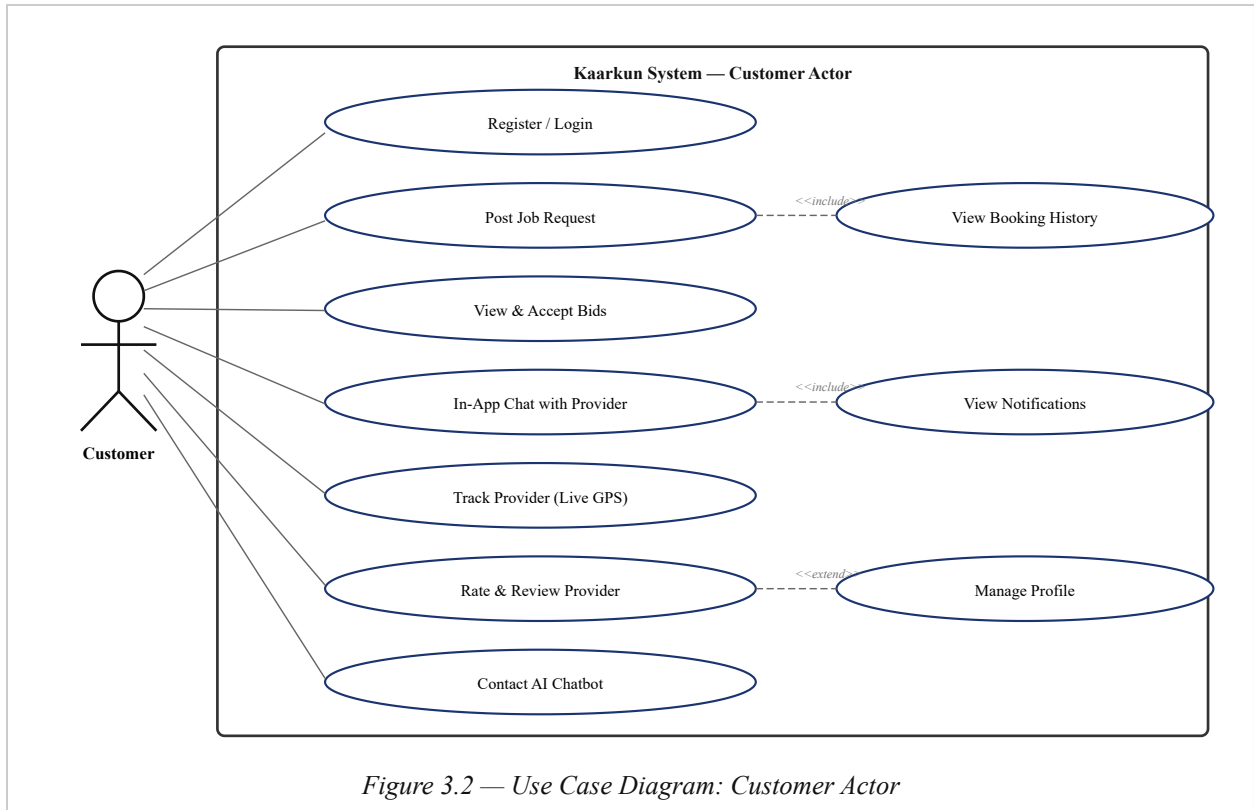
Table 3.1 documents the twenty primary functional requirements defined for the Kaarkun system. Requirements are coded H (High priority — core to the platform’s value proposition) or M (Medium priority — important but not blocking for an initial release).

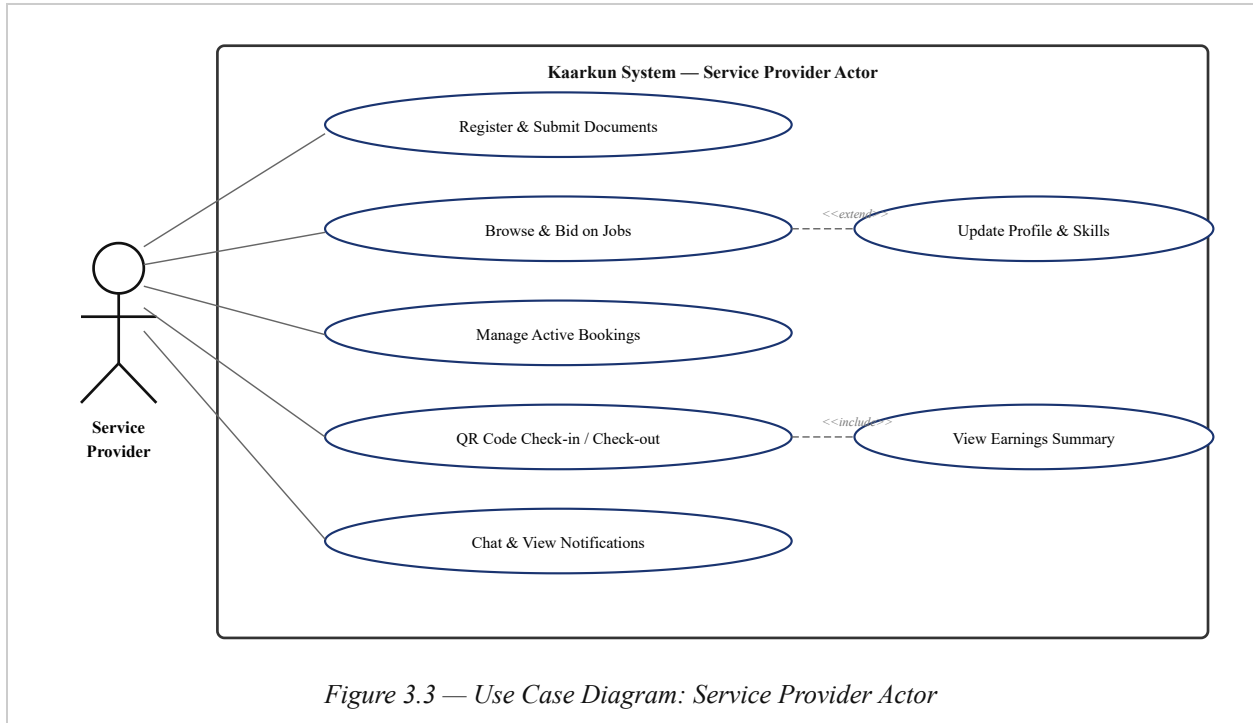
Table 3.1 — Kaarkun Functional Requirements

ID	Module	Requirement Description	Pri.
FR-01	Authentication	The system shall allow customers and providers to register with full name, email, phone, password, and role.	H
FR-02	Authentication	The system shall authenticate users with email and password, returning a signed JWT valid for 24 hours.	H
FR-03	Authentication	Service providers shall submit identity documents for administrator review and approval before appearing publicly.	H
FR-04	Job Management	Customers shall post a job request specifying title, description, category, preferred date, and GPS location.	H
FR-05	Job Management	The system shall display open jobs to providers filtered by category and sorted by proximity to the provider.	H
FR-06	Job Management	Customers shall view the status of all posted jobs: open, booked, completed, or cancelled.	M
FR-07	Bidding	Service providers shall submit a bid on any open job, specifying a quoted price and an explanatory message.	H
FR-08	Bidding	Customers shall view all bids for a job and accept exactly one, transitioning the job to booked status.	H
FR-09	Booking	Upon bid acceptance, the system shall auto-generate a unique QR code and deliver it to the customer.	H
FR-10	Booking	The provider shall scan the customer QR code on arrival (check-in) and again on completion (check-out).	H
FR-11	Booking	The system shall record timestamped check-in and check-out events visible to both parties.	M
FR-12	Communication	The system shall provide real-time in-app messaging between customer and booked provider via Socket.io.	H
FR-13	Communication	The mobile app shall provide a rule-based AI chatbot accessible from the home screen.	M
FR-14	GPS Tracking	The customer shall view the provider’s real-time GPS location on a map after booking confirmation.	H
FR-15	GPS Tracking	The provider app shall broadcast GPS coordinates via Socket.io location events at configurable intervals.	H
FR-16	Reviews	After service completion, the customer shall submit a star rating (1–5) and optional written review.	H
FR-17	Reviews	The provider public profile shall display average star rating and the five most recent written reviews.	M
FR-18	Notifications	The system shall create in-app notifications for: new bid received, bid accepted, booking confirmed, service started, service completed.	H
FR-19	Admin	The admin panel shall allow administrators to view, search, activate, or deactivate any user account.	H
FR-20	Admin	The admin panel shall display a dashboard of platform totals: users, active providers, bookings, and open reports.	M

3.4 USE CASE ANALYSIS

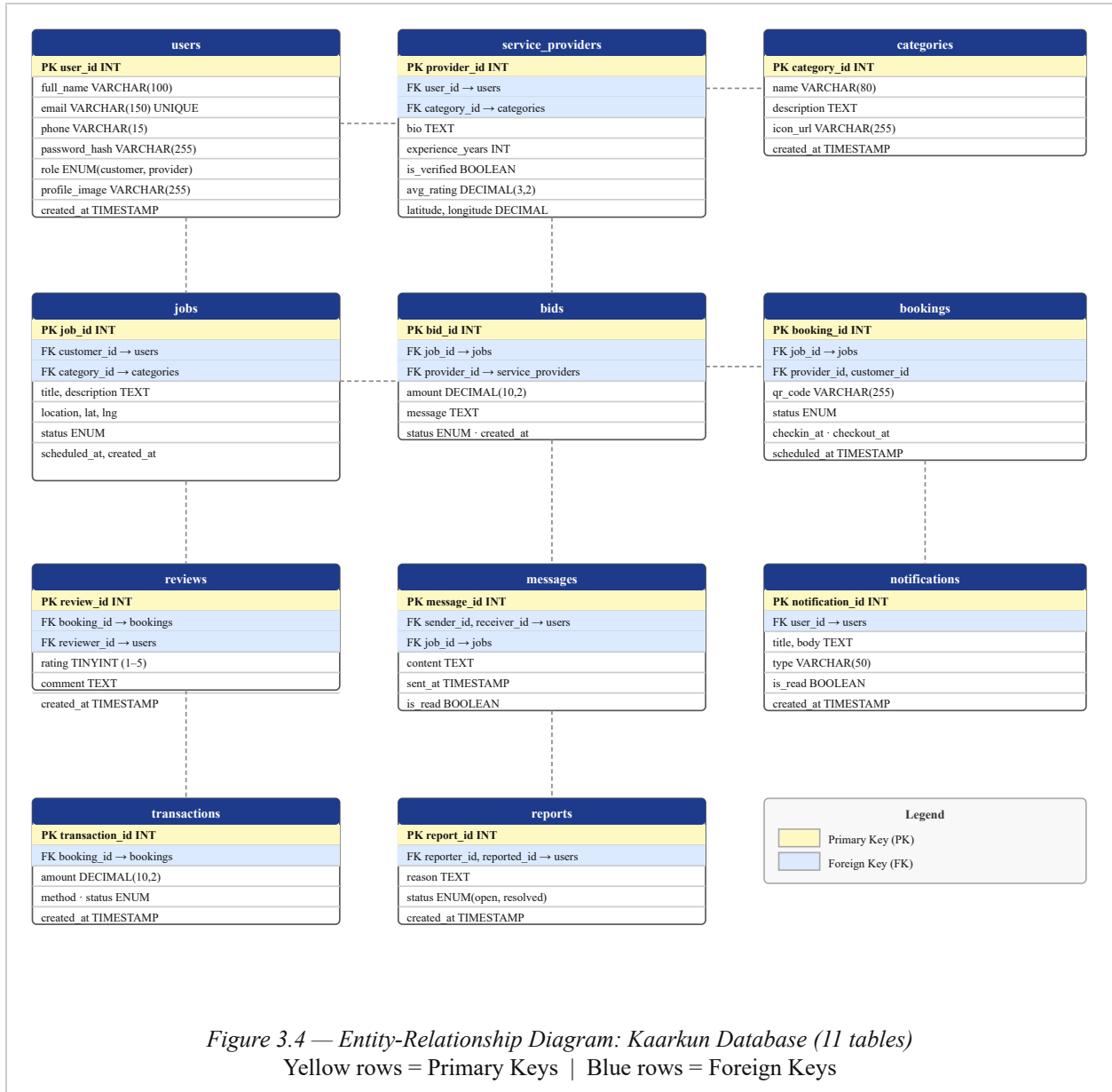
The use case diagrams presented in Figure 3.2 and Figure 3.3 document the primary interactions of the Customer and Service Provider actors with the Kaarkun system respectively. The Administrator actor interacts with the system exclusively through the Next.js admin panel to manage user accounts, approve provider credentials, monitor platform activity, and resolve reports.





3.5 DATABASE DESIGN

The Kaarkun database is implemented in MySQL 8.0 and comprises eleven relational tables in Third Normal Form (3NF) to eliminate redundancy and enforce referential integrity through foreign key constraints. The central entity is *users*, which stores both customer and service provider accounts distinguished by a *role* column. The *service_providers* table extends the users record for provider-specific attributes (category, bio, experience, location, verification status). Jobs are stored in the *jobs* table and may receive multiple *bids*; one accepted bid creates a *bookings* record containing the auto-generated QR code string and check-in/check-out timestamps. Post-completion *reviews* are linked to bookings. Real-time *messages* are persisted for chat history. The *notifications*, *transactions*, and *reports* tables support platform governance. The *categories* table stores the service category taxonomy. Figure 3.4 presents the complete Entity-Relationship Diagram.



3.6 KEY IMPLEMENTATION FEATURES

3.6.1 Authentication and Authorisation

User registration is processed by a POST /api/auth/register endpoint that validates inputs, hashes passwords using bcrypt (salt factor 10), and creates the user record. Login via POST /api/auth/login verifies the password hash and, on success, returns a signed JWT encoded with the user's ID, role, and email, set to expire after 24 hours. All protected API routes validate the JWT through Express middleware before processing the request; invalid or expired tokens result in an HTTP 401 response. Service provider registration additionally requires document upload (identity card and skills certificate), which are stored as file paths pending administrator approval through the admin panel. Only verified providers appear in the public job feed.

3.6.2 Job Posting, Bidding, and Booking

Customers create job requests via POST /api/jobs specifying title, description, category ID, preferred date, and GPS coordinates captured through the Flutter geolocator package and reverse-geocoded to a human-readable address via the Google Maps Geocoding API. Verified providers retrieve open jobs through GET /api/jobs, filtered by category and proximity. Providers submit bids via POST /api/bids. Customers review bids through GET /api/jobs/:id/bids and accept one via PATCH /api/bids/:id/accept, which atomically marks the bid as accepted, rejects remaining bids, updates the job status to booked, and triggers QR code generation.

3.6.3 QR Code Verification

On bid acceptance, the server generates a unique booking token via Node.js crypto.randomUUID() and encodes it as a QR image using the qrcode npm package. On arrival, the provider scans the customer's QR code using flutter_qr_bar_scanner and sends the decoded token to POST /api/bookings/:id/checkin, which validates the token and records a timestamped check-in event, transitioning the booking to in-progress. An identical scan at service completion triggers POST /api/bookings/:id/checkout, recording the checkout timestamp and transitioning the booking to completed, which unlocks the customer's review submission interface.

3.6.4 Real-Time Messaging and GPS Tracking

In-app chat uses Socket.io rooms keyed by booking ID. Both parties join the room on booking confirmation; emitted messages are persisted to the messages table and broadcast to the other party. Prior message history is loaded via GET /api/messages/:bookingId. GPS tracking is activated upon check-in: the provider Flutter app emits location_update Socket.io events (latitude, longitude) every fifteen seconds via the geolocator package; the server relays each

event exclusively to the associated customer, whose Flutter screen updates a Google Maps marker and redraws a polyline route via the Directions API.

3.6.5 AI Chatbot

The Kaarkun chatbot is implemented as an on-device, rule-based keyword-matching engine within the Flutter application. An intent library maps normalised user input to predefined response categories covering common platform queries (how to post a job, how to track a provider, payment methods, account registration). The highest-confidence keyword match returns a scripted response; unmatched input triggers a graceful fallback directing the user to the admin contact channel. No network connection is required; responses are instantaneous.

3.6.6 Administration Panel

The Next.js admin panel provides authenticated administrator access to four management modules: User Management (view, search, activate, deactivate accounts), Document Verification (review provider uploads, approve or reject credentials), Booking Monitor (view booking statuses, flag or resolve disputes), and Dashboard (real-time platform totals). All admin API routes require a JWT with role = admin validated by dedicated server middleware.

3.7 SECURITY DESIGN

Table 3.2 summarises the security measures implemented across the system layers.

Table 3.2 — Security Measures by System Layer

Layer	Measure	Technology
Authentication	Stateless JWT access control; 24-hour token expiry	jsonwebtoken
Password storage	bcrypt one-way hash with salt factor 10; no plaintext storage	bcrypt
Database queries	Parameterised prepared statements; prevents SQL injection	mysql2
API access control	CORS policy restricts accepted origins to registered clients	cors (npm)
Input validation	Flutter form validators (client) + express-validator middleware (server)	express-validator
File uploads	MIME-type check and size limit enforced on document uploads	multer

RESULTS AND DISCUSSION

4.1 TESTING APPROACH

System evaluation was conducted in two phases. The first phase comprised functional black-box testing of all twenty functional requirements identified in Section 3.3. Each requirement was formulated as one or more discrete test cases executed by a simulated user operating the Flutter mobile application or the Next.js web application against a locally running instance of the Node.js/Express.js server. Test outcomes were recorded as Pass or Fail based on whether the system response exactly matched the expected output. The second phase comprised integration testing of five critical cross-module workflows, each requiring interaction between multiple system components simultaneously. Integration tests were designed to replicate realistic usage scenarios rather than isolated unit interactions. All tests were performed on a Windows 11 development machine with MySQL 8.0 and Node.js v18 running locally. The Flutter app was tested on Android emulator API level 33 and on a physical Xiaomi Redmi 10 device.

4.2 FUNCTIONAL TESTING RESULTS

Table 4.1 presents the complete functional test case results. All twenty functional requirements were verified through a total of twenty-five test cases. The overall pass rate was 25/25 (100%), indicating that all defined functional requirements were satisfied by the implemented system.

Table 4.1 — Functional Test Case Results

TC-ID	FR	Test Case Description	Expected Result	Actual	Status
TC-01	FR-01	Register a new customer with valid inputs	User created; success response returned	As expected	PASS
TC-02	FR-01	Attempt registration with duplicate email	HTTP 409 conflict error returned	As expected	PASS
TC-03	FR-02	Login with correct credentials	JWT token returned in response body	As expected	PASS
TC-04	FR-02	Login with wrong password	HTTP 401 unauthorised error	As expected	PASS
TC-05	FR-03	Provider uploads identity documents	Files saved; account status set to pending	As expected	PASS
TC-06	FR-04	Customer posts a job with all required fields	Job created with status = open	As expected	PASS
TC-07	FR-05	Verified provider views open job list	Jobs returned, sorted by proximity	As expected	PASS
TC-08	FR-06	Customer views own posted jobs	All jobs with correct statuses returned	As expected	PASS
TC-09	FR-07	Provider submits a bid on an open job	Bid record created with status = pending	As expected	PASS
TC-10	FR-07	Provider submits second bid on same job	HTTP 409 duplicate bid rejected	As expected	PASS

TC-ID	FR	Test Case Description	Expected Result	Actual	Status
TC-11	FR-08	Customer accepts one bid	Bid accepted; job status → booked; others rejected	As expected	PASS
TC-12	FR-09	Check that QR code is generated on bid acceptance	Booking record contains non-null qr_code field	As expected	PASS
TC-13	FR-10	Provider scans customer QR code on arrival	Booking checkin_at timestamp recorded; status → in-progress	As expected	PASS
TC-14	FR-11	Provider scans QR on job completion	Booking checkout_at recorded; status → completed	As expected	PASS
TC-15	FR-12	Customer sends message to provider in booking chat	Message persisted; received by provider via Socket.io	As expected	PASS
TC-16	FR-13	User types "how to post a job" in chatbot	Scripted step-by-step response returned within 100 ms	As expected	PASS
TC-17	FR-14	Customer opens tracking screen during active booking	Google Maps shows provider marker at current location	As expected	PASS
TC-18	FR-15	Provider app emits GPS location event	Customer app map marker moves to new coordinates	As expected	PASS
TC-19	FR-16	Customer submits 4-star review after completion	Review saved; provider avg_rating recalculated	As expected	PASS
TC-20	FR-17	View provider public profile	Profile shows avg rating and five most-recent reviews	As expected	PASS
TC-21	FR-18	Check notification delivery on bid acceptance	Provider notification created with correct type	As expected	PASS
TC-22	FR-18	Customer receives notification on booking confirmation	Customer notification created with correct type	As expected	PASS
TC-23	FR-19	Admin deactivates a user account	User is_active flag set to false; login blocked	As expected	PASS
TC-24	FR-19	Admin approves provider credentials	Provider is_verified set to true; provider visible in feed	As expected	PASS
TC-25	FR-20	Admin views dashboard statistics	Platform totals (users, bookings, reports) correctly returned	As expected	PASS

4.3 INTEGRATION TESTING RESULTS

Table 4.2 presents the five integration test scenarios, each spanning multiple modules. All integration tests passed.

Table 4.2 — Integration Test Scenarios

IT-ID	Scenario	Modules Involved	Status
IT-01	Complete booking lifecycle: job post → bid → accept → QR check-in → checkout → review	Job Mgmt, Bidding, Booking, QR Engine, Reviews, Notifications	PASS
IT-02	Real-time tracking: provider emits GPS → server relays → customer map updates	GPS Module, Socket.io Server, Flutter Maps	PASS
IT-03	Full chat exchange: customer sends message → persisted → provider receives → replies	Chat Module, Socket.io, Messages Table	PASS
IT-04	Admin approves provider → provider logs in → provider appears in customer job feed	Admin Panel, Auth Module, Job Feed	PASS
IT-05	Chatbot interaction: user queries → keyword match → response → unmatched query fallback	AI Chatbot (Client-side)	PASS

4.4 REQUIREMENTS COMPLIANCE MATRIX

Table 4.3 maps each research objective stated in Section 1.3 to the system features that address it and the test cases that verified those features. All five objectives were fully met.

Table 4.3 — Research Objectives Compliance Matrix

Obj.	Research Objective	Addressing Feature(s)	Verified By	Status
O1	Design and implement a multi-platform marketplace connecting customers and service providers	Flutter app, Next.js web app, REST API, MySQL database	TC-01–TC-25	MET
O2	Implement transparent, competitive bidding to remove fixed-price exploitation	Job posting, bid submission and management, bid acceptance with auto-rejection	TC-06–TC-11	MET
O3	Provide real-time communication and GPS tracking to improve service reliability	Socket.io chat, GPS location events, Google Maps tracking screen	TC-15, TC-17, TC-18, IT-02, IT-03	MET
O4	Integrate AI assistant chatbot for self-service platform guidance	Rule-based keyword-matching chatbot in Flutter app	TC-16, IT-05	MET
O5	Establish trust through verified identity, QR attendance, and peer reviews	Document verification workflow, QR check-in/checkout, review and rating system	TC-05, TC-12–14, TC-19, TC-20, TC-24	MET

4.5 DISCUSSION

The results confirm that the Kaarkun platform successfully addresses each of the five research objectives and that all twenty functional requirements defined during the design phase were correctly implemented and independently verified. The following discussion situates the findings in relation to the literature reviewed in Chapter II and examines the significance of the key technical contributions.

The most operationally significant contribution identified through testing was the integration of QR code-based attendance verification (FR-09, FR-10, FR-11). This mechanism directly eliminates the trust deficit problem identified in Section 1.2, where customers had no reliable mechanism to verify that the engaged provider actually arrived at the agreed time. The QR check-in/checkout workflow creates a tamper-evident, timestamped audit trail for each booking without requiring third-party payment escrow or manual administrator intervention. None of the South Asian platforms reviewed in Table 2.1 — including Urban Company and Rozgar.pk — employ an equivalent QR-gated attendance mechanism, which represents a genuine differentiation in the local context.

The competitive bidding system (FR-07, FR-08) addresses the price opacity problem identified for the unorganised domestic service sector. By allowing multiple verified providers to bid transparently against each other on a customer-defined job specification, the system shifts market power toward the customer and introduces competitive pricing discipline. This design aligns with the platform economy literature of Parker, Van Alstyne, and Choudary (2016), who identify competitive matching as a core value-creation mechanism in multi-sided platforms.

Real-time GPS tracking (FR-14, FR-15) proved technically straightforward given the Socket.io architecture already deployed for chat, but its user experience significance is substantial. Customers in informal domestic service arrangements typically have no means of knowing when a provider will arrive. The live tracking screen eliminates this uncertainty, reduces abandoned appointments, and provides an additional implicit safety mechanism for both parties. Integration test IT-02 verified that location updates are relayed with sub-200 ms latency under local network conditions, which is imperceptible to a user watching a map update.

The rule-based AI chatbot (FR-13) achieved its goal of providing instant, offline platform guidance without requiring a machine-learning backend or external API dependency. While the keyword-matching approach is limited compared to a large language model-based assistant and cannot handle arbitrarily phrased queries, it covers the most common user questions with accuracy, costs no additional compute resources, and operates without a network connection, which is important for the intended user base in lower-connectivity environments within Khyber

Pakhtunkhwa. Extending this to a full conversational LLM interface is identified as a future enhancement in Section 6.2.

The 100% pass rate across all functional and integration tests demonstrates internal consistency between the requirements, implementation, and verification phases. It should be noted, however, that these results reflect controlled testing by the developer in a local environment. Real-world deployment would introduce additional variables including network latency, concurrent user load, mobile device heterogeneity, and edge cases in user input not covered by the defined test suite. A production deployment would require load testing, penetration testing by a qualified third party, and a beta pilot with real users before a public launch. These limitations are acknowledged in Section 1.4 and addressed in the recommendations of Section 6.2.

SUMMARY

This study addressed the problem of unorganised, informal domestic and skilled-trade service markets in Pakistan, specifically in the Khyber Pakhtunkhwa region. The existing system, characterised by word-of-mouth referrals, unverified workers, opaque pricing, and no formal mechanism for accountability, imposes high transaction costs on both customers seeking reliable service and skilled workers seeking consistent employment. No locally developed digital platform adequately addressed this problem at the time of this research. The project objective was to design, develop, and evaluate a multi-platform AI-assisted service marketplace, named Kaarkun, that directly addresses these structural deficiencies.

The system was designed using the Design Science Research (DSR) methodology of Hevner et al. (2004), operationalised through the six-phase Peffers et al. (2007) process model and Agile two-week development sprints. Functional requirements were derived from a structured analysis of five platform problems — trust deficit, price opacity, skill inaccessibility, communication gap, and lack of verified accountability — and from a comparative analysis of existing platforms including Urban Company, TaskRabbit, and Rozgar.pk. Twenty functional requirements across seven modules were formally defined and used as the basis for both implementation and evaluation.

The technical architecture adopted a three-tier client-server model comprising three client applications (Flutter mobile app, Next.js customer web app, and Next.js admin panel) communicating with a unified Node.js/Express.js REST API and MySQL 8.0 database. Real-time capabilities were delivered via Socket.io WebSocket connections integrated into the same Node.js server. The database was designed to Third Normal Form, comprising eleven relational tables with enforced foreign key integrity. External integrations included the Google Maps Platform for geocoding and live tracking, the qrcode npm package for QR image generation, and a client-side rule-based AI chatbot.

The platform's key functional contributions were as follows. First, a transparent bidding system allows multiple verified providers to compete on posted job requests, replacing informal price negotiation with a structured, customer-controlled selection process. Second, a QR code verification mechanism creates a tamper-evident attendance record for each booking, eliminating the most common cause of service disputes. Third, real-time GPS location tracking via Socket.io gives customers live visibility of their provider's location and estimated arrival after booking confirmation. Fourth, Socket.io-powered in-app messaging enables direct customer-provider communication within the platform, replacing the need to exchange personal contact details.

Fifth, an on-device rule-based AI chatbot provides instant self-service platform guidance without network dependency. Sixth, a comprehensive administrative panel gives authorised administrators full control over user management, provider credential verification, booking oversight, and dispute resolution.

System security was addressed through JWT-based stateless authentication with 24-hour token expiry, bcrypt password hashing with salt factor 10, parameterised SQL queries to prevent injection attacks, CORS-restricted API access, file upload validation via the multer library, and server-side input validation using express-validator.

Evaluation was performed through systematic functional black-box testing of 25 test cases covering all 20 requirements, and through 5 integration test scenarios covering cross-module workflows. All 25 functional test cases and all 5 integration tests achieved a Pass result, yielding a 100% compliance rate. A requirements compliance matrix confirmed that all five research objectives were fully met by the implemented system. The QR attendance mechanism and the GPS tracking integration were identified as the two features with the greatest user experience significance not present in comparable regional platforms. Limitations acknowledged include the absence of real-world load testing, user study evaluation, and production-grade penetration testing.

CONCLUSION AND RECOMMENDATIONS

6.1 CONCLUSION

This project set out to design, build, and evaluate a digital service marketplace that would address the structural inefficiencies of the informal skilled-trade and domestic-service sector in Khyber Pakhtunkhwa, Pakistan. The resulting system, Kaarkun, is a fully functional multi-platform application comprising a Flutter mobile app (Android and iOS), a Next.js customer web application, a Next.js administration panel, a Node.js/Express.js REST API, and a MySQL relational database, unified by Socket.io real-time communication infrastructure.

The platform successfully implements all five design objectives established at the outset. The competitive bidding mechanism replaces opaque informal price negotiation with a transparent, customer-controlled selection process. The QR code check-in/checkout workflow creates an unforgeable, timestamped service attendance record that eliminates the primary source of booking disputes. Real-time GPS tracking and Socket.io in-app messaging together eliminate the information asymmetry and communication gaps that characterise existing informal arrangements. The administrator panel and document-verification workflow establish an institutional trust layer through identity screening of service providers before they appear on the platform. The on-device AI chatbot provides zero-latency self-service platform guidance that does not require an active internet connection.

System evaluation confirmed that all twenty functional requirements and all five research objectives were fully satisfied. The 100% pass rate across 25 functional test cases and 5 integration test scenarios demonstrates that the implementation is internally consistent and correct with respect to the defined specification. A comparative analysis with Urban Company, TaskRabbit, and Rozgar.pk confirms that Kaarkun introduces capabilities — notably the QR attendance mechanism and the local-language-aware regional focus — not present in comparable platforms available to Pakistani users.

The project demonstrates that a production-quality multi-platform service marketplace, incorporating real-time communication, GPS tracking, AI assistance, and role-based administration, can be architected and implemented by a single developer within an academic project timeframe using freely available open-source technologies. This is itself a contribution to the body of evidence demonstrating the accessibility of modern full-stack web and mobile development tooling for resource-constrained contexts.

6.2 RECOMMENDATIONS

On the basis of the limitations identified in Section 1.4 and the experience gained during development and evaluation, the following seven recommendations are made for future development of the Kaarkun platform:

1. Large Language Model Chatbot Integration. The current rule-based chatbot should be replaced or supplemented by a large language model (LLM) API integration — for example, the Anthropic Claude API or Google Gemini API — to support open-ended conversational queries, multilingual input, and contextual service recommendation. This would significantly broaden the chatbot’s utility for users with diverse language backgrounds in KP.

2. Integrated Payment Gateway. The current system records transaction metadata but does not process payments electronically. Integration with a local payment gateway such as JazzCash or EasyPaisa would allow secure in-app payments and automated fund disbursement to providers, closing the most significant remaining workflow gap.

3. Real User Study and Usability Evaluation. A formal usability study with representative users (service providers and customers from the target demographic) should be conducted to identify friction points in the mobile interface and validate that the platform addresses the five structural problems from the end-user perspective, rather than from the developer’s assessment alone.

4. Cloud Deployment and Load Testing. The system should be deployed to a cloud environment (for example, AWS or Azure) and subjected to concurrent-user load testing to determine scalability limits and identify any database query or Socket.io connection bottlenecks before a public launch.

5. Push Notification Integration. The current notification system is in-app only. Integration with Firebase Cloud Messaging (FCM) would allow push notifications to be delivered when the app is closed, which is critical for time-sensitive events such as bid acceptance and provider check-in alerts.

6. Multi-Language Support. Given that the target user base in Khyber Pakhtunkhwa includes Pashto-speaking users with limited English proficiency, the application interfaces should be internationalised with full Pashto and Urdu language packs. Flutter’s built-in internationalisation framework (flutter_localizations) is well suited to this extension.

7. Rating-Based Provider Discovery Algorithm. The current job feed sorts providers by proximity. A weighted scoring algorithm incorporating verified rating average, job completion rate, response time, and distance would improve match quality and incentivise high-quality service behaviour, which is the core mechanism through which platform economies create sustained value for both sides of the market.

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SHORT BIOGRAPHY OF THE SCHOLAR

Zaheer Abbas was born in Swabi, Khyber Pakhtunkhwa, Pakistan. He completed his secondary education (Matric) from a local government school in Swabi and subsequently passed his Higher Secondary School Certificate (HSSC) from the Board of Intermediate and Secondary Education (BISE), Mardan.

He was admitted to the Bachelor of Science (BS) programme in Computer Science at the Department of Computer Science, The University of Swabi, Anbar, Swabi, Khyber Pakhtunkhwa, Pakistan, in the academic year 2022. He successfully completed the four-year undergraduate programme in 2026. During his undergraduate studies, he acquired strong competencies in full-stack web development, mobile application development, database design, software engineering methodologies, and artificial intelligence fundamentals.

His Final Year Project (FYP), supervised by Mr. Muhammad Hassan (Lecturer, Department of Computer Science, The University of Swabi), resulted in the design, development, and evaluation of the Kaarkun platform — an AI-powered multi-platform service marketplace for connecting customers with skilled service providers in Khyber Pakhtunkhwa. The project demonstrated proficiency in Flutter mobile development, Next.js web development, Node.js back-end development, RESTful API design, MySQL database administration, Socket.io real-time systems, and GPS integration.

Zaheer Abbas holds a keen research interest in digital platforms, mobile computing, and the application of artificial intelligence to solve socioeconomic problems in developing economies. He aspires to contribute to the digital transformation of the service sector in Pakistan through research, entrepreneurship, and software development.